

# SHUVAM PAL

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## Education

**Dayananda Sagar College of Engineering**

**Bengaluru, India**

*B.E. Artificial Intelligence and Machine Learning*

## Experience

**Bentham AI** 

*Dec 2025 – Feb 2026*

*Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions*

*Remote*

- Built a resilient **Puppeteer/Node.js workflow** for MCA filings with session handling, CAPTCHA handoff, form retries, and state recovery, reducing manual filing time by **85%**.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized backend services with **Docker**, wired **GitHub Actions** deployments to **Google Cloud Run**, and improved release velocity by **70%**.

**Google Summer of Code** 

*May 2025 – Sept 2025*

*Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

*Remote*

- Built medical-image segmentation tooling for **95 brain subparts**, handling model inference, orientation, preprocessing, and large-volume data paths with reproducible outputs.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Worked in a mature open-source codebase with review constraints, compatibility concerns, and careful integration into existing user workflows.

## Open Source

*[Contributions](#) *

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside existing maintainer security, reliability, and review expectations.

## Projects

**Murdoc** | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

**[GitHub](#)** 

- Built a self-hosted **AI security gateway** for LLM, HTTP-tool, and MCP traffic with route profiles, request scoring, allow/block policies, and runtime checks before downstream API execution.
- Blocked **90%+** prompt-injection and unsafe-tool attempts across a **150-case attack corpus** with false positives under **5%**; added RBAC, PII redaction, and audit ledgers for **100%** of policy decisions.

**Postificus** | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

**[GitHub](#)** 

- Built Go/Echo backend services for a **content distribution platform** using browser automation, with REST APIs, RabbitMQ workers, Redis autosave, and PostgreSQL persistence.
- Added DLQs, retries, circuit-breaker isolation, and failure telemetry so unreliable external API/browser workflows stayed recoverable instead of cascading through distributed workers.

**Seaweed: Contest Platform** | *Go, PostgreSQL, Redis, Judge0, AWS, Docker, Kubernetes*

**[GitHub](#)** 

- Built **Go backend services** for registration, submissions, sandboxed code execution, scoring, and live contest rankings across **500+ concurrent users**.
- Designed async Judge0 dispatch, PostgreSQL result schemas, Redis-backed runtime state, and Docker/Kubernetes deployment on AWS, achieving **sub-5s P95 verdict latency**.

## Technical Skills

**Languages:** Go, JavaScript/TypeScript, Python

**Frameworks:** Echo, FastAPI, Gin, Express, GraphQL

**Tools:** PostgreSQL, Redis, RabbitMQ, Docker, GitHub Actions, Prometheus, OpenTelemetry, GCP, AWS